

Quaternion-Based Attitude Tracking Control Design for UAVs

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Abstract—This paper utilizes quaternions to represent the attitude of UAVs and proposes an attitude controller. We use backstepping methodology together with fractional exponent to achieve salient performance in tracking a desired attitude trajectory. Numerical simulation results are provided to verify the presented scheme and demonstrate its effectiveness.

Keywords—Quadrotor UAVs, Quaternion, Attitude Control

I. INTRODUCTION

In recent years, drone technology has made significant strides across various industries, demonstrating exceptional capabilities and potential in fields such as military applications, entertainment, photography, and geographical exploration. As technology rapidly advances, drones have evolved from mere tools to becoming a pivotal technology that is reshaping modern industrial models and lifestyles. The versatility of drones allows them to adapt to different scenarios, effectively solving real-world problems and enhancing efficiency. With the ongoing maturation of artificial intelligence (AI) and automation technologies, drones have become an indispensable core technology in numerous fields.

This research is motivated by a desire to delve into the fundamental operational principles and technical details of drones to better understand their flight capabilities, maneuverability, and adaptability in various environments. By examining these aspects, we aim to uncover the potential applications of drones across different industries. Through analyzing real-world use cases of drones in various sectors, we can gain a clearer understanding of their specific benefits in terms of improving work efficiency, reducing operational costs, and providing innovative solutions. Moreover, with the advancement of AI algorithms, the application of drones in autonomy and intelligence has become a crucial research focus. This study not only aims to explore the potential of drones in various fields by gaining an in-depth understanding of their basic operations but also aspires to drive further innovation and development in this domain, thereby bringing more advanced and practical technological solutions to society.

In this study, we have chosen quaternions as the foundational computational tool. By leveraging the mathematical properties of quaternions, we can describe the motion trajectories of drones with greater precision, enabling more efficient navigation and control. The application of this method not only enhances the operational performance of drones but also lays a solid foundation for future technological innovations.

Drones have become a transformative force across many sectors, offering new ways to tackle traditional challenges. In the military, drones have revolutionized surveillance, reconnaissance, and targeted strikes, providing precise and timely information while minimizing human risk. The entertainment industry has also benefited from drone technology, with aerial footage providing dynamic perspectives in

filmmaking and live events that were previously unattainable. Photographers and videographers have embraced drones as essential tools, allowing them to capture stunning aerial views and creative shots that add a new dimension to visual storytelling.

In geographical exploration, drones have opened up new possibilities for surveying and mapping. They can access remote and difficult-to-reach areas, collecting data quickly and accurately. This capability is particularly valuable in environmental monitoring, disaster management, and infrastructure inspection, where traditional methods may be less effective or more dangerous.

The integration of AI into drone technology is driving the development of autonomous systems that can perform complex tasks with minimal human intervention. AI-powered drones can analyze their surroundings in real-time, make decisions, and adjust their flight paths accordingly. This autonomy is essential for applications such as search and rescue operations, where speed and precision are critical. Additionally, AI enhances the drones' ability to perform tasks like object detection, tracking, and data analysis, making them more efficient and reliable tools for various industries.

The future of drone technology is closely tied to the continued advancement of AI and automation. As these technologies evolve, drones are expected to become even more intelligent, capable of performing an increasingly wide range of tasks autonomously. This evolution will likely lead to the development of new applications and industries centered around drone technology, further integrating them into our daily lives and work processes.

In conclusion, drones have rapidly evolved from simple remote-controlled devices to sophisticated tools with a wide array of applications [1], [2]. Their versatility and adaptability make them invaluable in many industries, and ongoing research into their capabilities and potential will only expand their utility. By understanding the fundamental principles of drone operation and exploring the latest technological advancements, we can continue to innovate and push the boundaries of what drones can achieve, ultimately contributing to a future where drones play a central role in solving some of the most pressing challenges facing society. Many people use quaternions combined with PD control to design the necessary controls for drones [3], [4]. In previous research papers, some have also utilized LQR (Linear Quadratic Regulator) to design the required controls for drones [5].

One can visualize the basic coordinate system of the drone through Figure 1 [6]. The drone itself has three rotational directions, which correspond to the three Euler angles: yaw (ψ), roll (ϕ), and pitch (θ). These rotations are formed by rotating around the X, Y, and Z axes of a right-handed Cartesian coordinate system.

In paper [7], we can learn about two types of backstepping control designs for UAV attitude tracking under the influence of

uncertainties and external disturbances. The quaternion-based attitude stabilization backstepping procedure is extended to the attitude tracking control design. It demonstrates the efficiency and performance of the designed control.

In paper [8], a novel nonlinear model predictive control (NMPC) was used for navigation and obstacle avoidance. Different types of trajectories were categorized, enabling the system to predict future obstacle positions. This system is used to handle dynamic obstacles in real time.

Paper [9] proposes a robust position control method, using integral sliding mode and backstepping sliding mode controllers as control strategies, and analyzes their stability and performance through Lyapunov stability analysis.

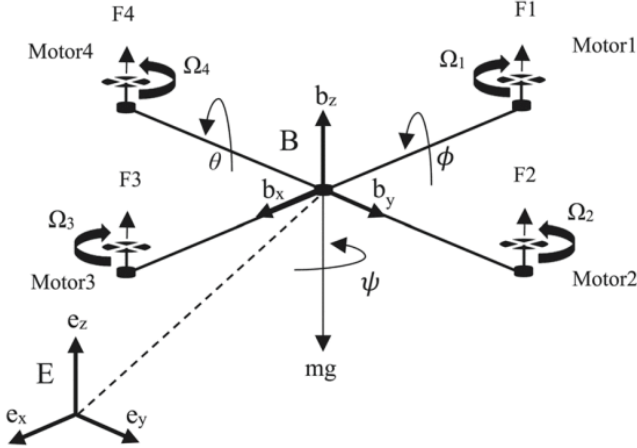


Fig. 1. Attitude dynamics of quadrotor [6]

II. PRELIMINARIES FOR QUATERNIONS

The purpose of this research is to gain an in-depth understanding of drone motion and, through the modeling of motion equations, achieve a stable flight state, with data uploaded to a simulation platform for further analysis. Before proceeding, it is essential to understand the mathematical tools we are using. Why do we use quaternions for control? The reasons are as follows: quaternions can avoid gimbal lock, they can represent orientation with only four numbers, and they allow for smooth interpolation. However, their drawback is also evident—we cannot intuitively visualize the orientation. In the paper [1], we can learn how to effectively use quaternions, gaining a basic understanding of them, which will facilitate our subsequent calculations. The paper [2] provides a clearer insight into how we can achieve the desired control.

$$q = q_0 + i q_1 + j q_2 + k q_3 \quad (1)$$

In equation (1), q_0 represents the magnitude of the vector, while q_1 , q_2 , and q_3 are the vectors in the three directions of the quaternion. The matrix representation is shown in equation (2).

$$q = [q_0 \ q_1 \ q_2 \ q_3]^T \quad (2)$$

The conjugate of a quaternion is defined similarly to that of a regular complex number. We define q^* as the conjugate of a quaternion, and its definition is given by equation (3).

$$\text{Conj}(q) = q^* = [q_0 \ -q_1 \ -q_2 \ -q_3]^T \quad (3)$$

The length of a quaternion is similar to that of ordinary complex numbers. We define the length of a quaternion as $\|q\|$, which is given by the formula in equation (4).

$$\text{Norm}(q) = \|q\| = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2} \quad (4)$$

In the article, we will also use the inverse of a quaternion, denoted as q^{-1} , which is defined as shown in Equation (5).

$$\text{Inv}(q) = q^{-1} = \frac{q^*}{\|q\|^2} \quad (5)$$

For ease of calculation, we need to normalize the quaternion. We define the normalization of the quaternion as $\text{Normalize}(q)$, which is defined in Equation (6).

$$\text{Normalize}(q) = \frac{q}{\|q\|} \quad (6)$$

Assume we have two quaternions, p and q . In quaternion operations, multiplication is not commutative, implying $p \otimes q \neq q \otimes p$. The multiplication of quaternions is shown in Equation (7)

$$p \otimes q = \begin{bmatrix} p_0 q_0 - p_1 q_1 - p_2 q_2 - p_3 q_3 \\ p_0 q_1 + p_1 q_0 + p_2 q_3 - p_3 q_2 \\ p_0 q_2 - p_1 q_3 + p_2 q_0 + p_3 q_1 \\ p_0 q_3 + p_1 q_2 - p_2 q_1 + p_3 q_0 \end{bmatrix} \quad (7)$$

Note that both $p \otimes p^{-1}$ and $p \otimes p^*$ can yield $[1; 0; 0; 0]$. Although quaternions are not commutative, this does not affect the result in current context.

III. ATTITUDE DYNAMICS OF UAVS

After the introduction to quaternions, we can begin designing our drone attitude control system. Before the design, we need to understand the attitude model of the drone. The model is shown as follows, where Equation (8) represents the quaternion differential equation.

$$\dot{q} = \frac{1}{2} q \otimes \begin{bmatrix} 0 \\ \omega \end{bmatrix} \quad (8)$$

From Equation (9), we can determine the torque magnitude required in the drone's attitude model.

$$\dot{\omega} = J^{-1}[\tau - \omega \times (J\omega)] \quad (9)$$

IV. CONTROLLER DESIGN

A. Error Functions

After understanding the kinematic model, we can use the known equations to design the required torque. Before doing so, we need to define the error. The quaternion error is shown in Equation (10). When there is no error, $q_{err} = [1; 0; 0; 0]$.

$$q_{err} = q_{ref}^* \otimes q \quad (10)$$

The error in ω is defined as in Equation (11)

$$\omega_{err} = \omega - \omega_{ref} \quad (11)$$

B. Controller Design

After knowing the error, we need to determine the error dynamics. The quaternion error dynamics can be derived from Equation (8), and the error dynamics are shown in Equation (12). These can be obtained using Equations (10) and (11).

$$\dot{q}_{err} = \frac{1}{2} q_{err} \otimes \begin{bmatrix} 0 \\ \omega_{err} \end{bmatrix} \quad (12)$$

To ensure the convergence of equation (12), we design an ideal ω_{err} represented as $\omega_{err,d}$. The designed ω_{err} is shown in equation (13), and k_1 must be greater than zero.

$$\begin{bmatrix} 0 \\ \omega_{err,d} \end{bmatrix} = -k_1 q_{err}^* \otimes (q_{err} - \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}) \quad (13)$$

When we substitute Equation (13) back into Equation (12), we can obtain Equation (14), which will cause Equation (12) to converge.

$$\dot{q}_{err} = \frac{-k_1}{2} (q_{err} - \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}) \quad (14)$$

The error dynamics of ω are shown in Equation (15).

$$\dot{\omega}_{err} = \dot{\omega} - \dot{\omega}_{ref} \quad (15)$$

Since we want to design the control through ω_{err} , and this design cannot be directly controlled, we refer to the designed control as $\omega_{err,d}$. Therefore, we also need another error, which we denote as δ , as shown in Equation (16).

$$\delta = \omega_{err} - \omega_{err,d} \quad (16)$$

We can rewrite equation (12) from equation (16) as equation (17).

$$\dot{q}_{err} = \frac{-k_1}{2} (q_{err} - \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}) + \frac{1}{2} q_{err} \otimes \begin{bmatrix} 0 \\ \delta \end{bmatrix} \quad (17)$$

The error dynamics of δ are shown in Equation (18). We can first substitute into Equation (15) and then use Equation (9) to obtain Equation (19). In Equation (19), we can identify the required torque.

$$\dot{\delta} = \dot{\omega}_{err} - \dot{\omega}_{err,d} \quad (18)$$

$$\dot{\delta} = J^{-1}[\tau - \omega \times (J\omega)] - \dot{\omega}_{ref} - \dot{\omega}_{err,d} \quad (19)$$

The term $\dot{\omega}_{err,d}$ in Equation (19) can be obtained from Equation (8) and can be substituted into Equation (20).

$$\begin{bmatrix} 0 \\ \dot{\omega}_{err,d} \end{bmatrix} = \frac{d}{dt} \left\{ -k_1 q_{err}^* \otimes (q_{err} - \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}) \right\} \quad (20)$$

Since we expect δ to converge to 0, we set $\tau = \tau_1 + \tau_2$. We expect τ_1 to eliminate all components of δ in the dynamics, which gives us Equation (21). τ_2 is the term that ensures δ converges, as shown in Equation (22). In Equation (22), k_2 must be greater than zero, and its value can control the rate of convergence.

$$\tau_1 = \omega \times (J\omega) + J\dot{\omega}_{ref} + J\dot{\omega}_{err,d} \quad (21)$$

$$\tau_2 = -k_2 J \delta^{1/3} \quad (22)$$

Next, we substitute equations (21) and (22) back into equation (19), and we obtain equation (23), as shown below. From this, we can see that equation (19) will converge.

$$\dot{\delta} = -k_2 \delta^{1/3} \quad (23)$$

Using the above formulas, we can create the required attitude control system. The control schematic is shown in Figure 2, where we can see the relationships between each part, which helps clarify the connections between the components used.

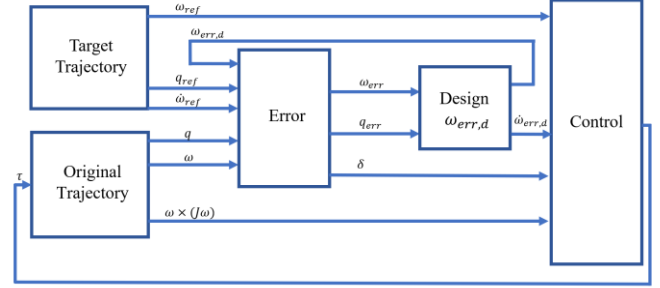


Fig. 2. A schematic of proposed attitude control system

V. SIMULATION RESULTS

We set ω_{ref} as the target to be tracked, which consists of three sine waves. Before we understand all the necessary equations and the attitude control diagram in Figure 1, we need to first identify all the parameters we are using. The parameters are as follows: set the moment of inertia $J = \text{diag}[0.1; 0.1; 0.12]$, set the constants $k_1 = 20$ 、 $k_2 = 2$. We arbitrarily define the trajectory to be tracked. The initial values for the following figures are set as $\omega_{ref}(0) = [-0.1; -0.2; 0.2]$ and $q_{ref}(0) = [0.94628; -0.1541; 0.19051; -0.21098]$, while the initial values for the drone are: $\omega(0) = [-0.1; -0.2; 0.2]$ and $q(0) = [0.94628; -0.1541; 0.19051; -0.21098]$.

Figure 3 shows q_{err} and from the figure we can see that q_{err} eventually converges to $[1; 0; 0; 0]$ indicating that q and q_{ref} have aligned. For convenience, we convert the quaternions q and q_{ref} into attitude angles in degrees. This allows us to observe the yaw, pitch, and roll angles, which are shown in Figures 4, 5, and 6, respectively.

Figure 7 shows that ω_{err} converges to zero at around 0.8 seconds. We separately plot ω and ω_{ref} for roll, pitch, and yaw motions as shown in Figures 8, 9, and 10, respectively. One can clearly see that they eventually converge to the same values.

Figure 11 shows that δ converges to zero in about 0.4 seconds.

As for the torque, it converges to zero when equilibrium is reached, as shown in Figure 12.

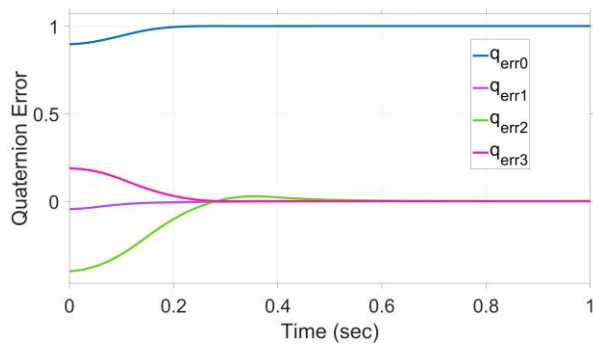


Fig. 3. Trajectory of quaternion error

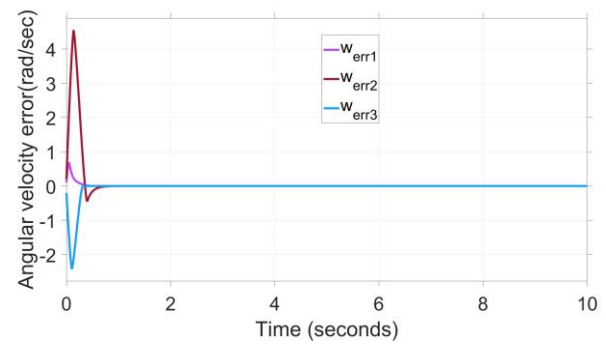


Fig. 7. Trajectory of angular velocity error

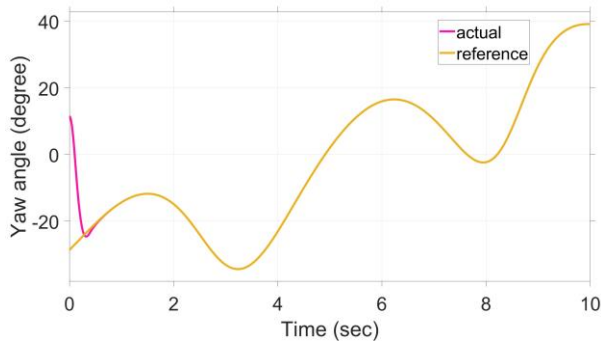


Fig. 4. Trajectory of yaw angle tracking

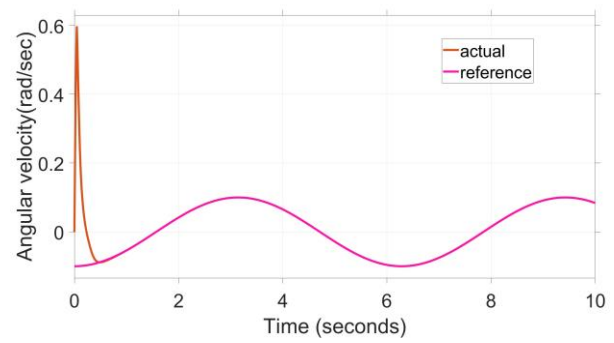


Fig. 8. Trajectory of roll angular velocity tracking

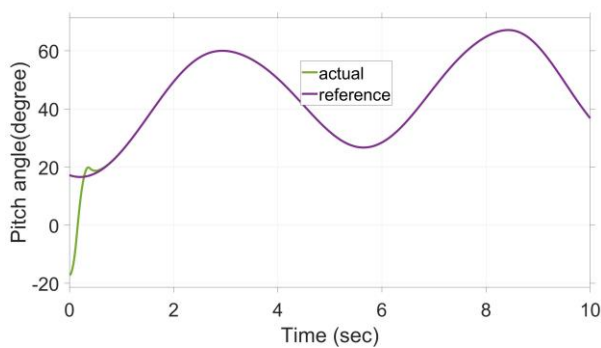


Fig. 5. Trajectory of pitch angle tracking

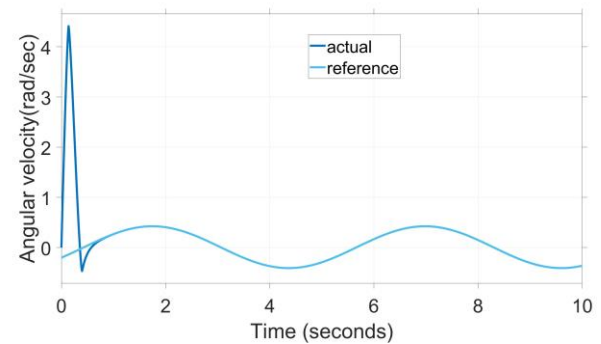


Fig. 9. Trajectory of pitch angular velocity tracking

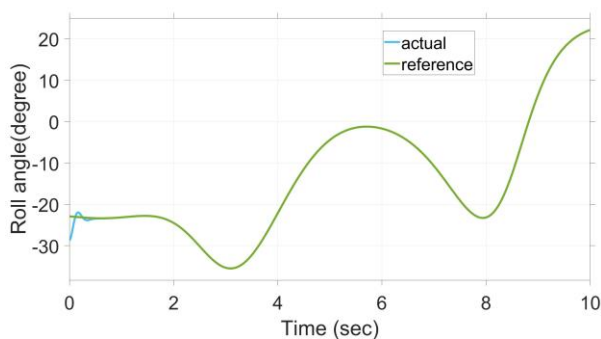


Fig. 6. Trajectory of roll angle tracking

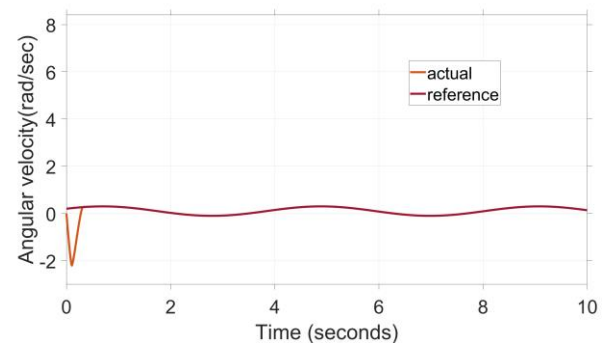


Fig. 10. Trajectory of yaw angular velocity tracking

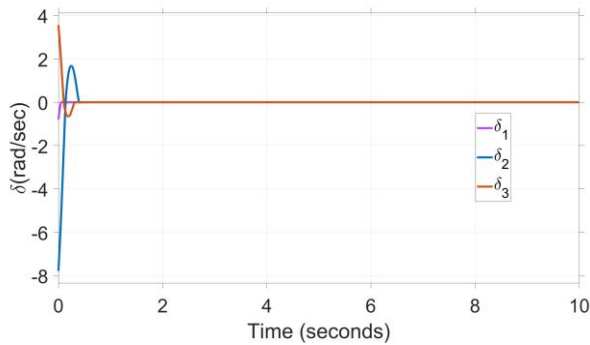
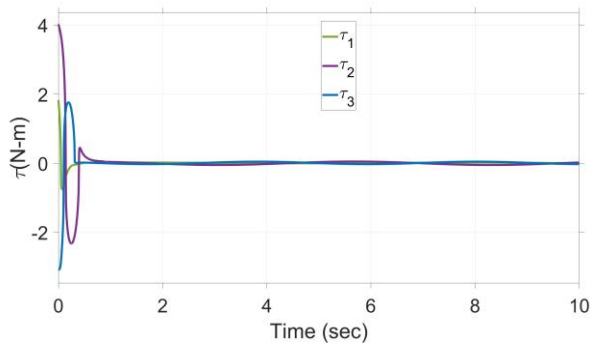
Fig. 11. Trajectory of δ 

Fig. 12. Trajectory of torque

VI. CONCLUSIONS

This paper used quaternions to represent the attitude of UAVs and employed the backstepping technique to design an attitude tracking controller ensuring all errors associated with trajectory tracking would converge to zero in a short period. The presented simulation results validated our design and supported its effectiveness and feasibility. In the future, we will conduct experiments to assess the performance of the proposed scheme for further improvement.

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